Chapter 12

Money, Banking, Prices, and Monetary Policy

Learning Objectives

12.1 Explain the functions of money, and how money is measured.

12.2 Construct the monetary intertemporal model.

12.3 Derive the Fisher relation.

12.4 Construct a competitive equilibrium in the monetary intertemporal model and carry out equilibrium experiments using the model.

12.5 Demonstrate that money is neutral in the monetary intertemporal model.

12.6 List the factors that can shift money demand and show how a shift in money demand affects economic variables in the monetary intertemporal model.

12.7 Show how conventional monetary policy is ineffective in a liquidity trap and explain unconventional monetary policies.

What is Money?

- Medium of exchange
- Store of value
- Unit of account

Measures of Money

- Monetary Base (outside money) = currency in circulation + bank reserves
- M1 = currency in circulation + transactions deposits + travelers' checks + demand deposits
- M2 = M1 + savings deposits + retail money market funds

The Inflation Rate

* $$i = \frac{P' - P}{P}$$

The Fisher Relation

$$1 + r = \frac{1 + R^*}{1 + i}$$

The Approximate Fisher Relation

$$r^* \approx R - i$$

Figure 12.1 The Nominal Interest Rate vs. Inflation

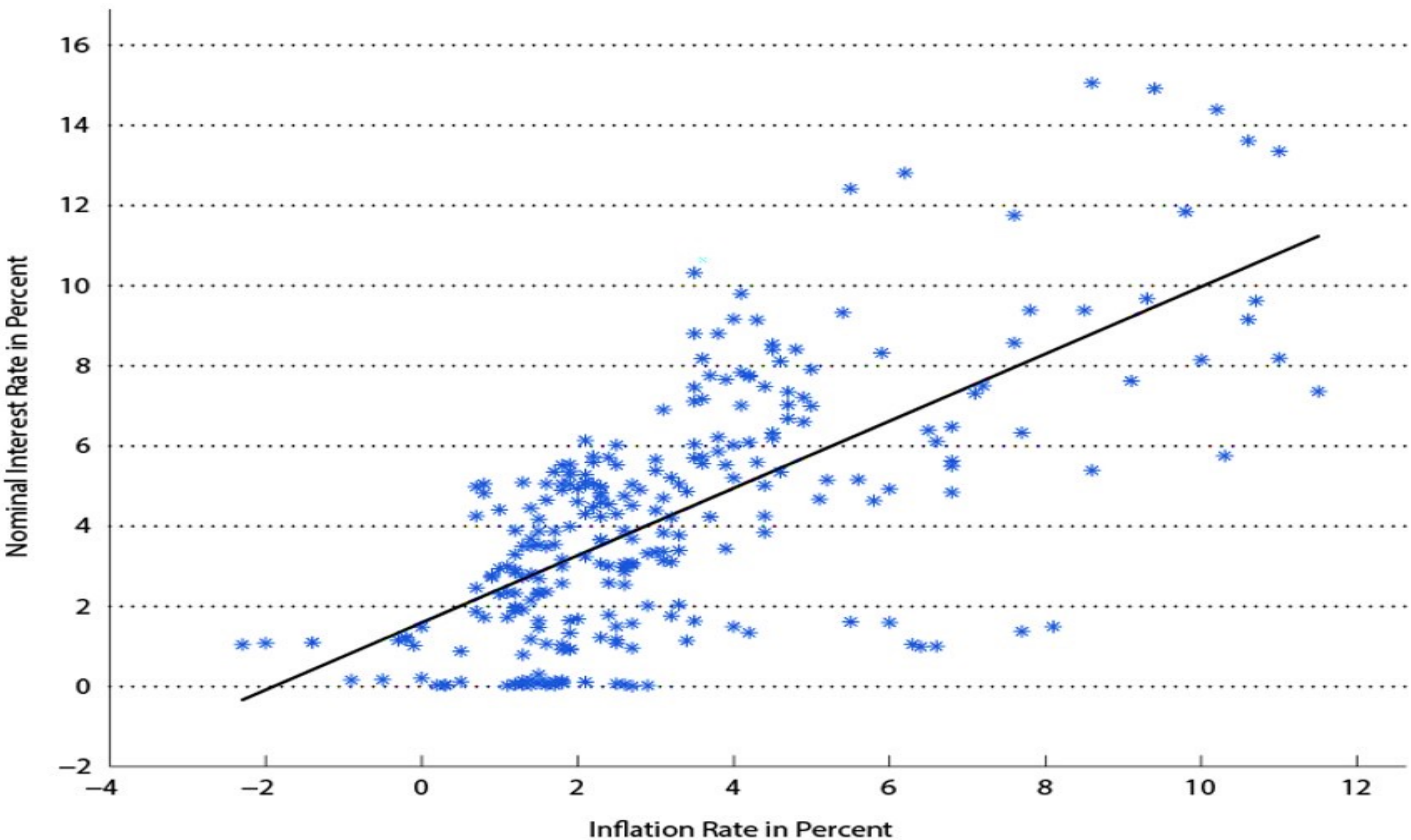
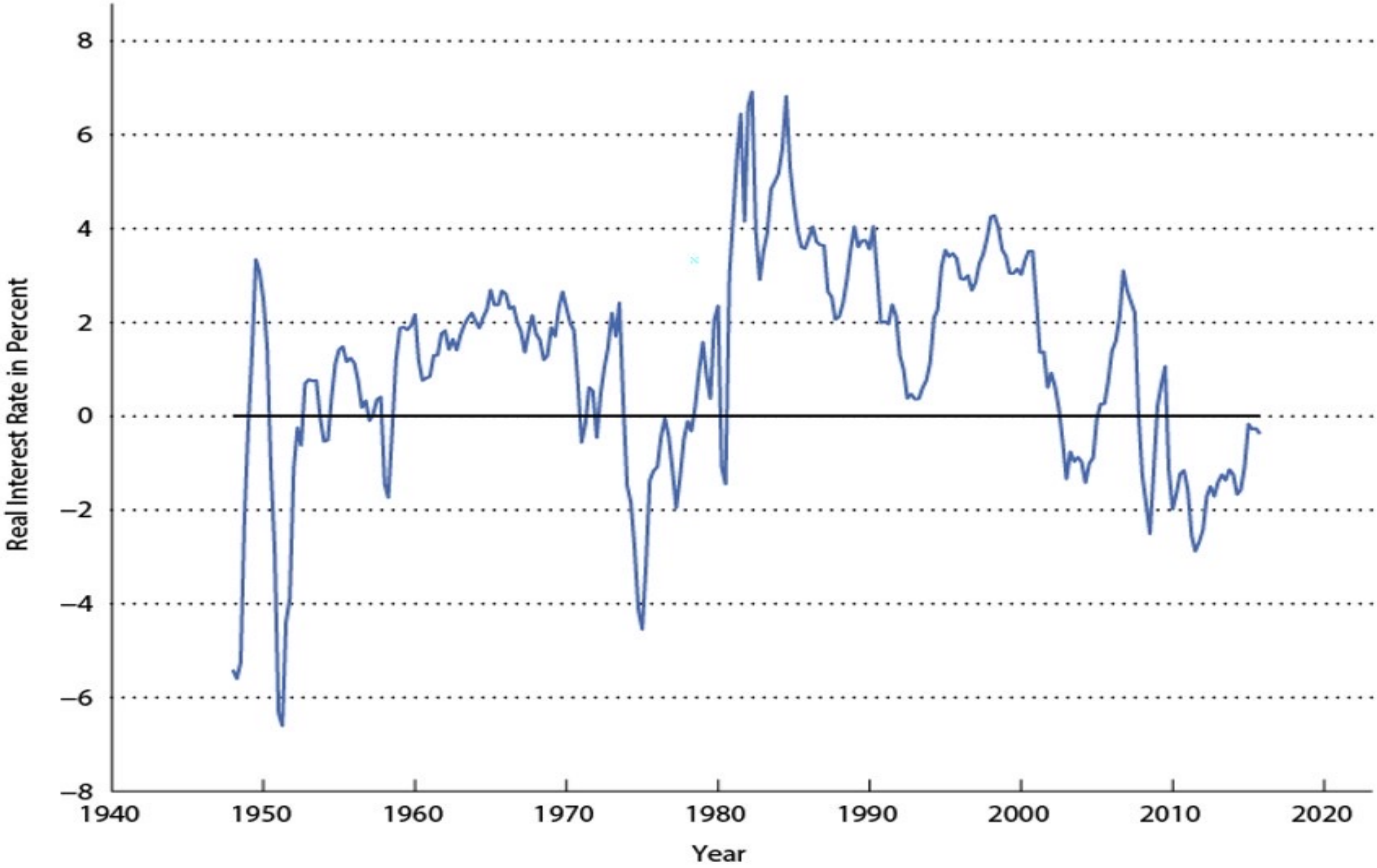


Figure 12.2 The Measured Real Interest Rate



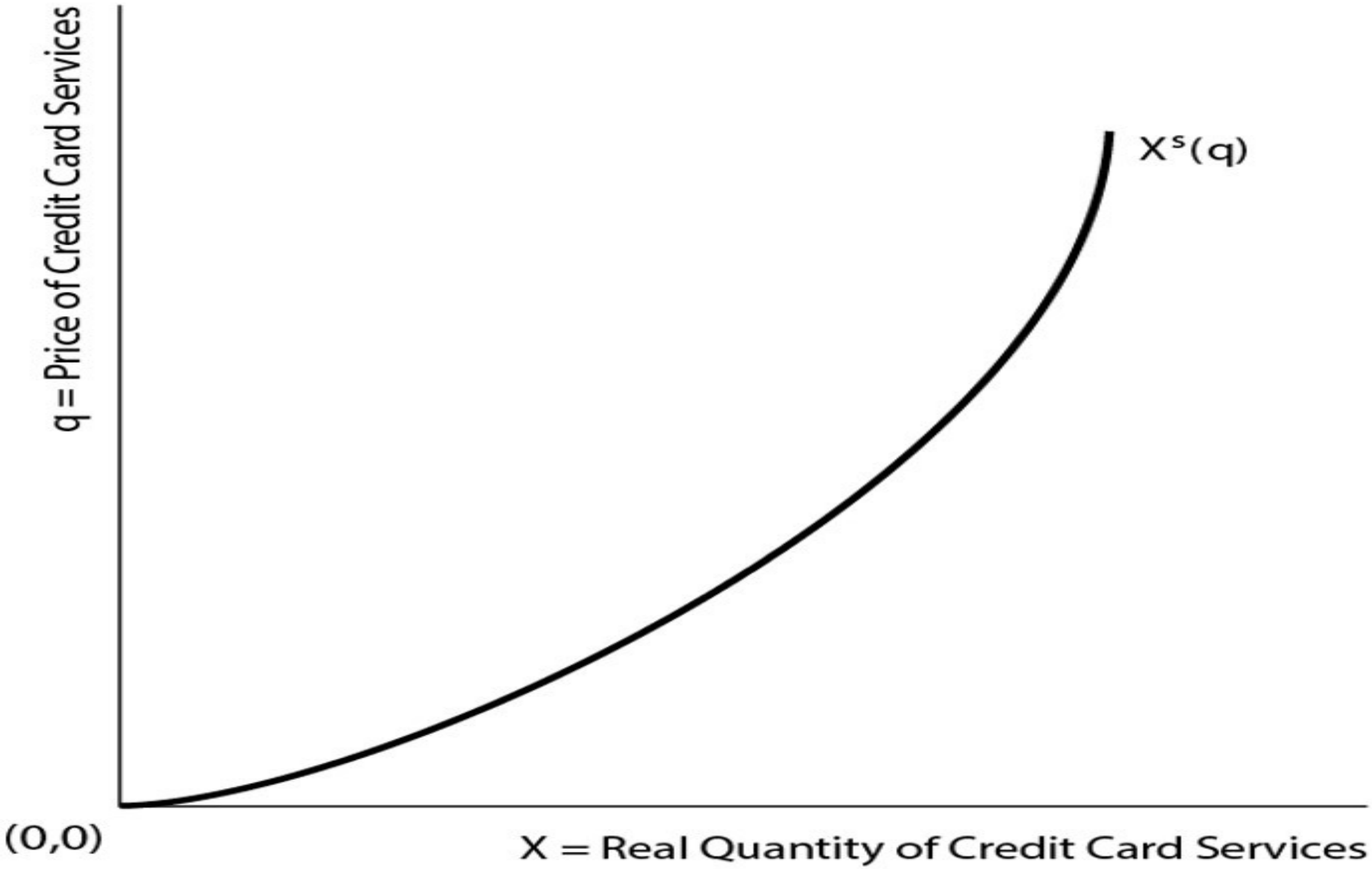
Banks and Alternative Means of Payment (1 of 2)

- Assume all goods must be purchased with currency or credit cards.
- "Credit cards" can be assumed to stand in, more broadly, for debt cards and checks, for example – all alternative means of payment supplied by the financial system.
- Goods purchased at price *P*, no matter what means of payment is used.

Banks and Alternative Means of Payment (2 of 2)

- Using a credit card costs q per unit of goods purchased.
- Credit supply (by banks) is given by $X^s(q)$, which is increasing in q.
- Supplying credit card services is costly for banks.

Figure 12.3 The Supply Curve for Credit Card Services



Demand for Credit Card Services

- Quantity of goods purchased with credit card services: $X^d(q)$
- Goods purchased with currency: $Y - X^d(q)$
- If $P(1+R) > P(1+q)$,

then all goods are purchased with credit cards.

- If $P(1+R) < P(1+q)$,

then all goods are purchased with currency.

Figure 12.4 Equilibrium in the Market for Credit Card Services

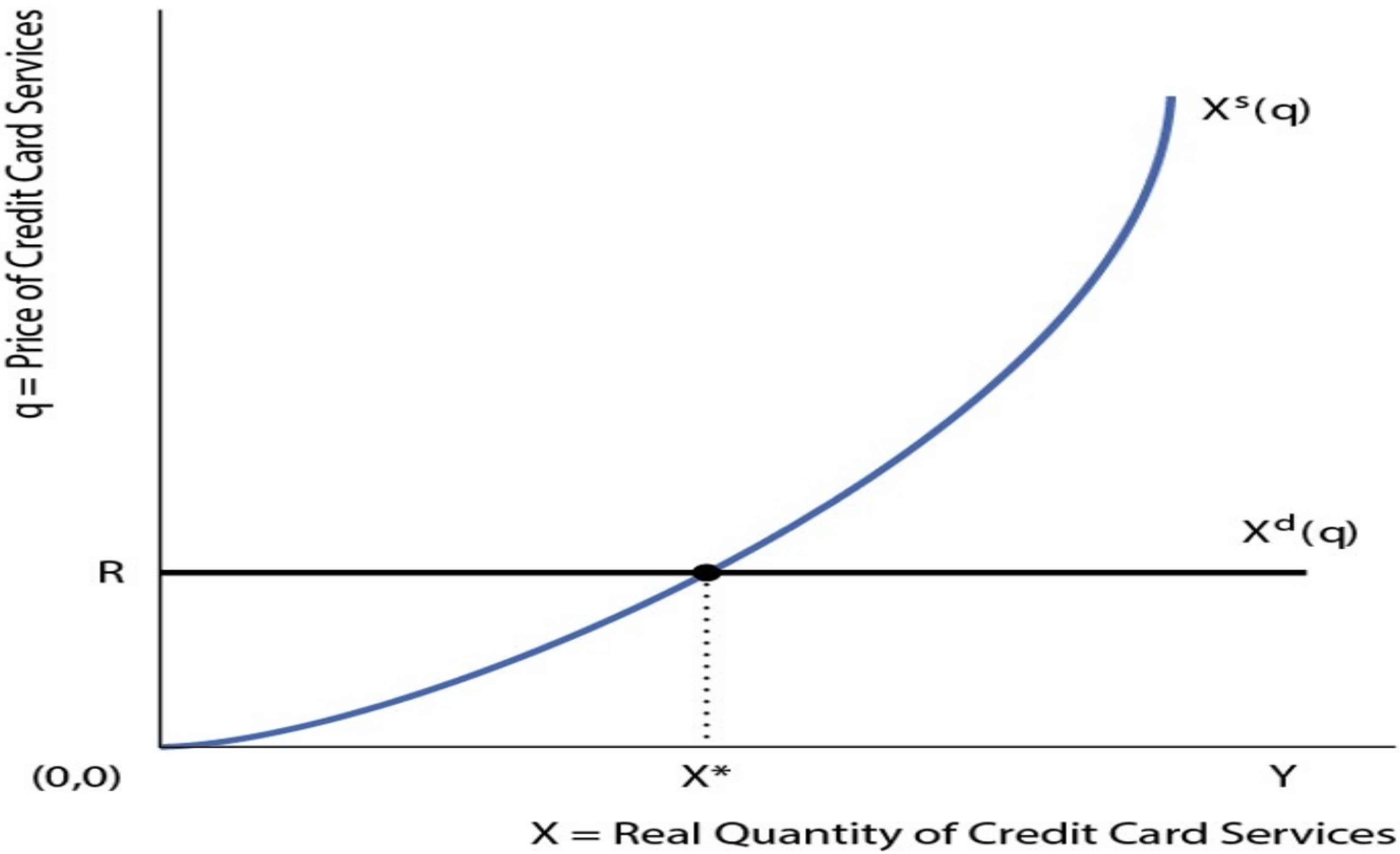
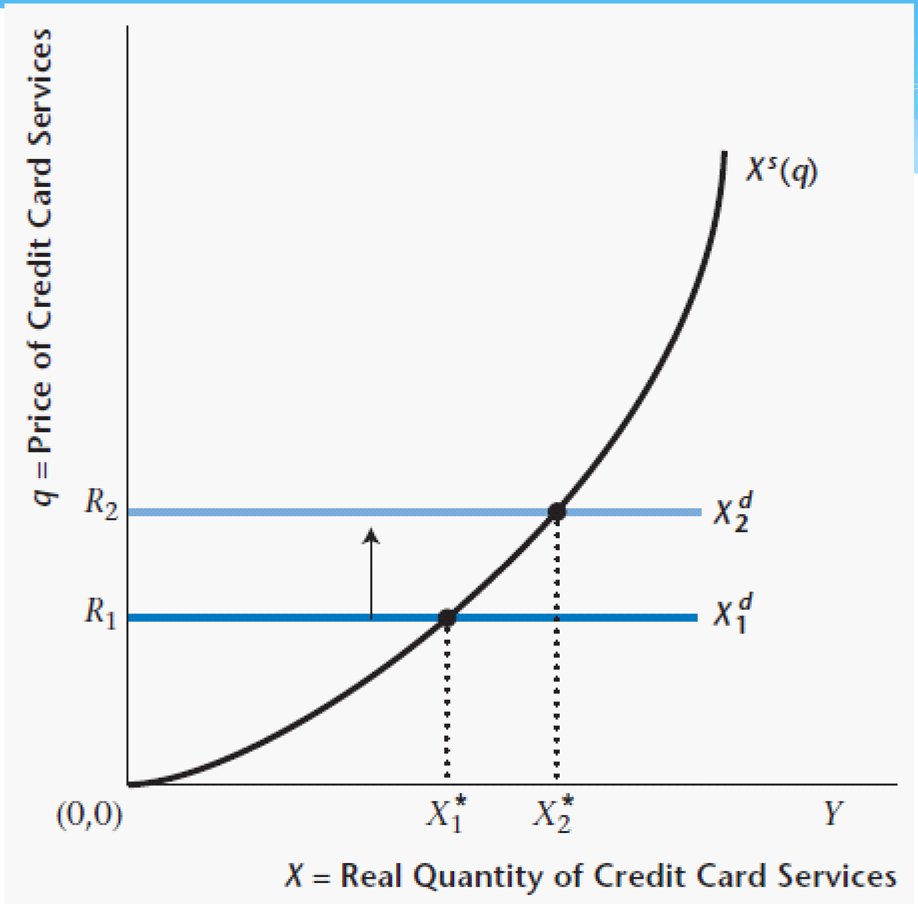


Figure 12.5 Increase in the Nominal Interest Rate and the Market for Credit Card Services



Demand for Money

* $$M^d = P[Y - X^*(R)]$$
* Here, $X^*(R)$ is the equilibrium quantity of credit card services (decreasing function of R). Therefore, more simply:

$$M^d = PL(Y, R)$$

Demand for Money (2 of 2)

* Increasing in real income – more currency required as volume of transactions increases.
* Decreasing in the nominal interest rate. The nominal interest rate is the opportunity cost of using currency in transactions – higher R implies greater use of credit in transactions, and less use of currency.

Nominal Money Demand

* Substitute using the approximate Fisher relation:

$$M^d = PL(Y, r + i)$$

* For our experiments, suppose inflation rate is zero (harmless):

$$M^d = PL(Y, r)$$

Figure 12.6 The Nominal Money Demand Curve in the Monetary Intertemporal Model

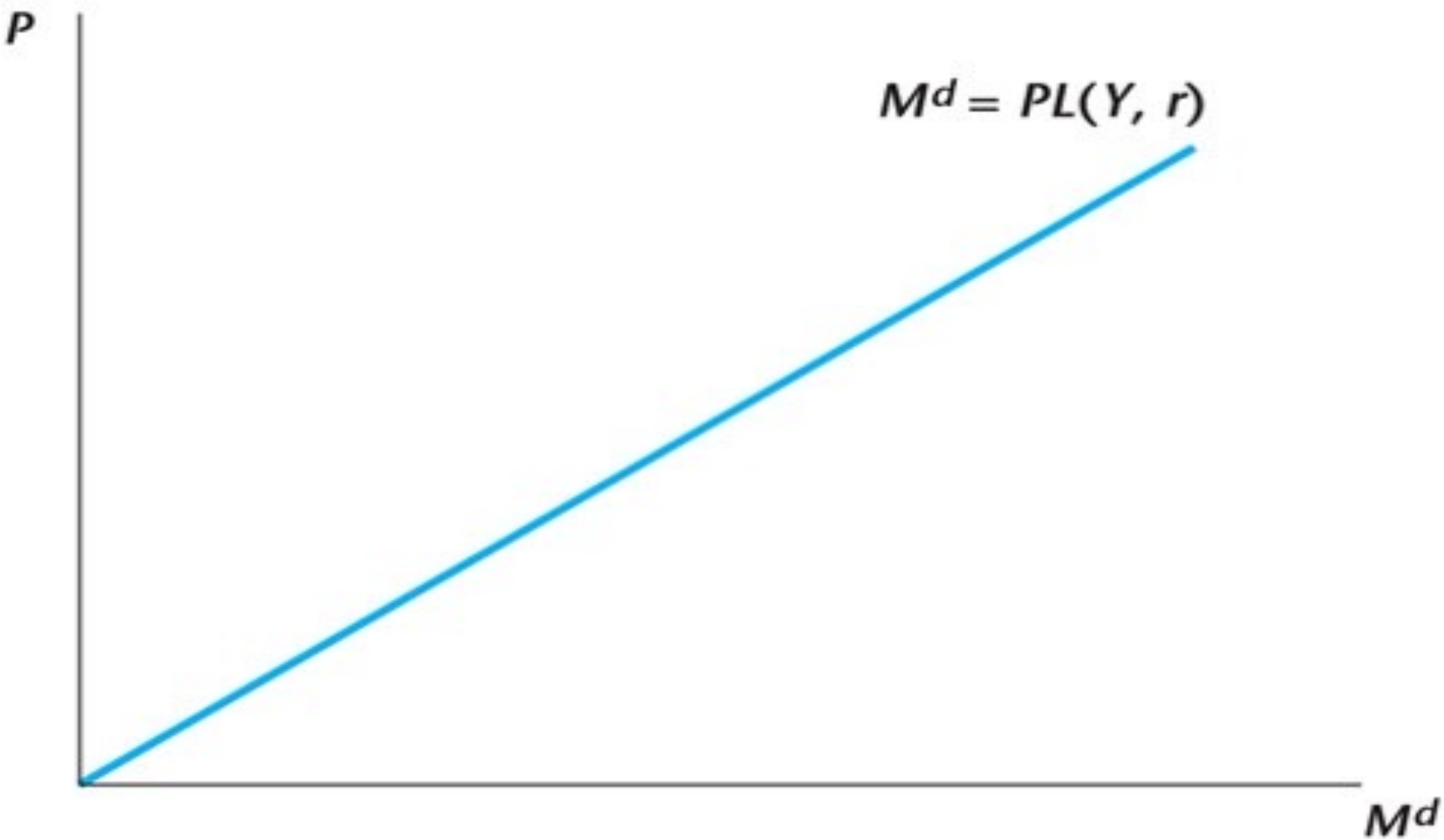


Figure 12.7 Increase in Current Real Income and the Nominal Money Demand Curve

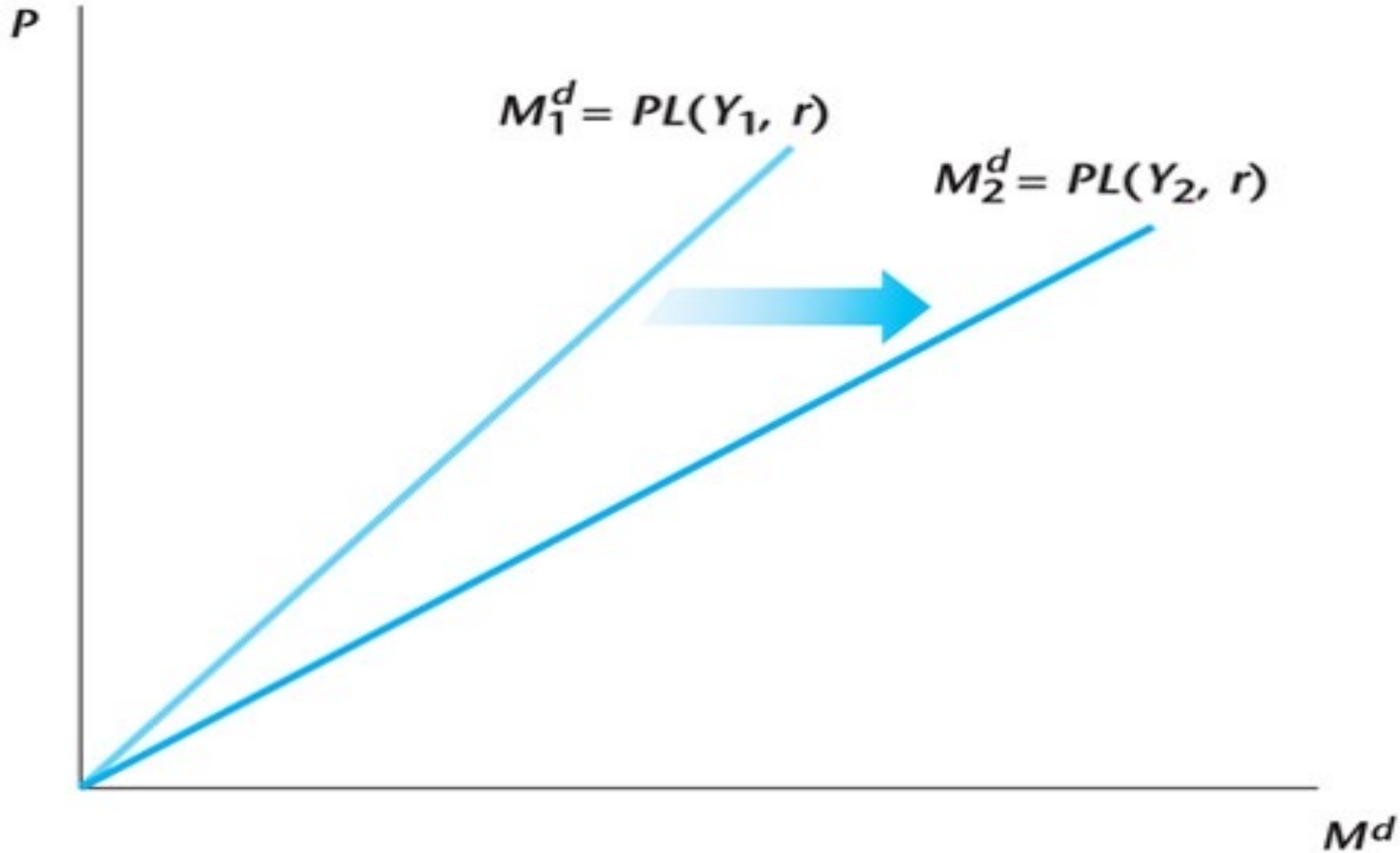


Figure 12.8 The Current Money Market in the Monetary Intertemporal Model

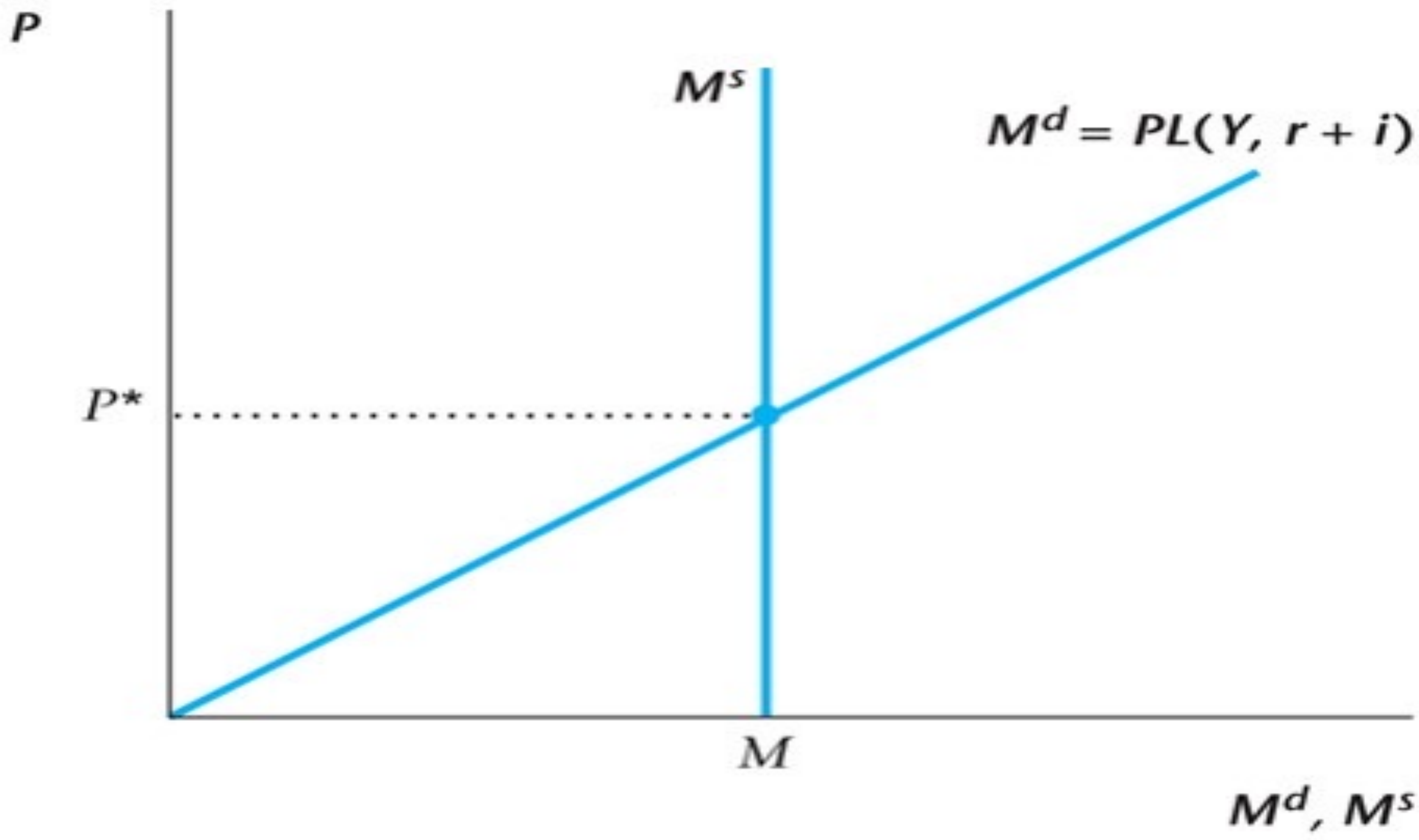


Figure 12.9 The Complete Monetary Intertemporal Model

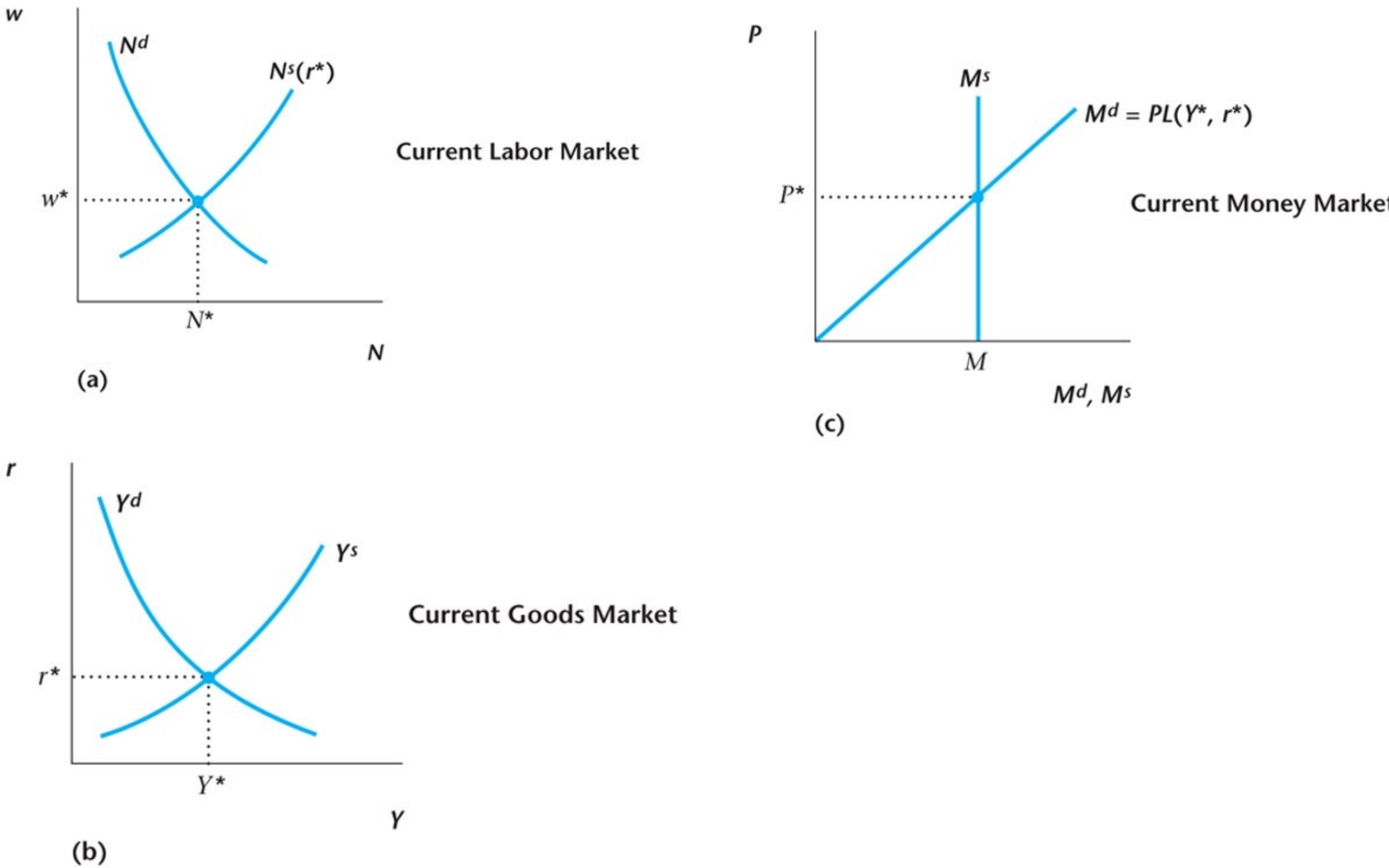
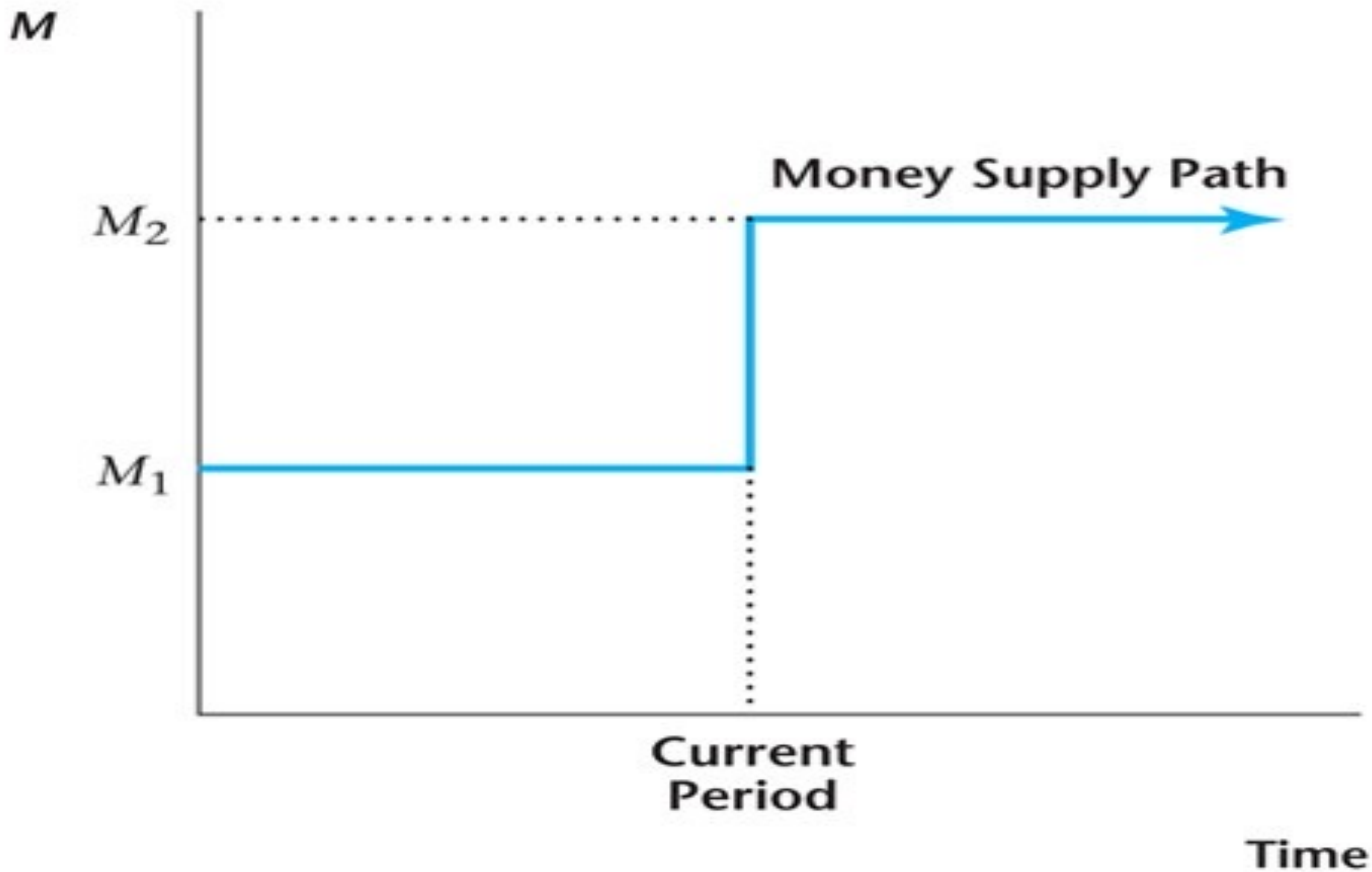


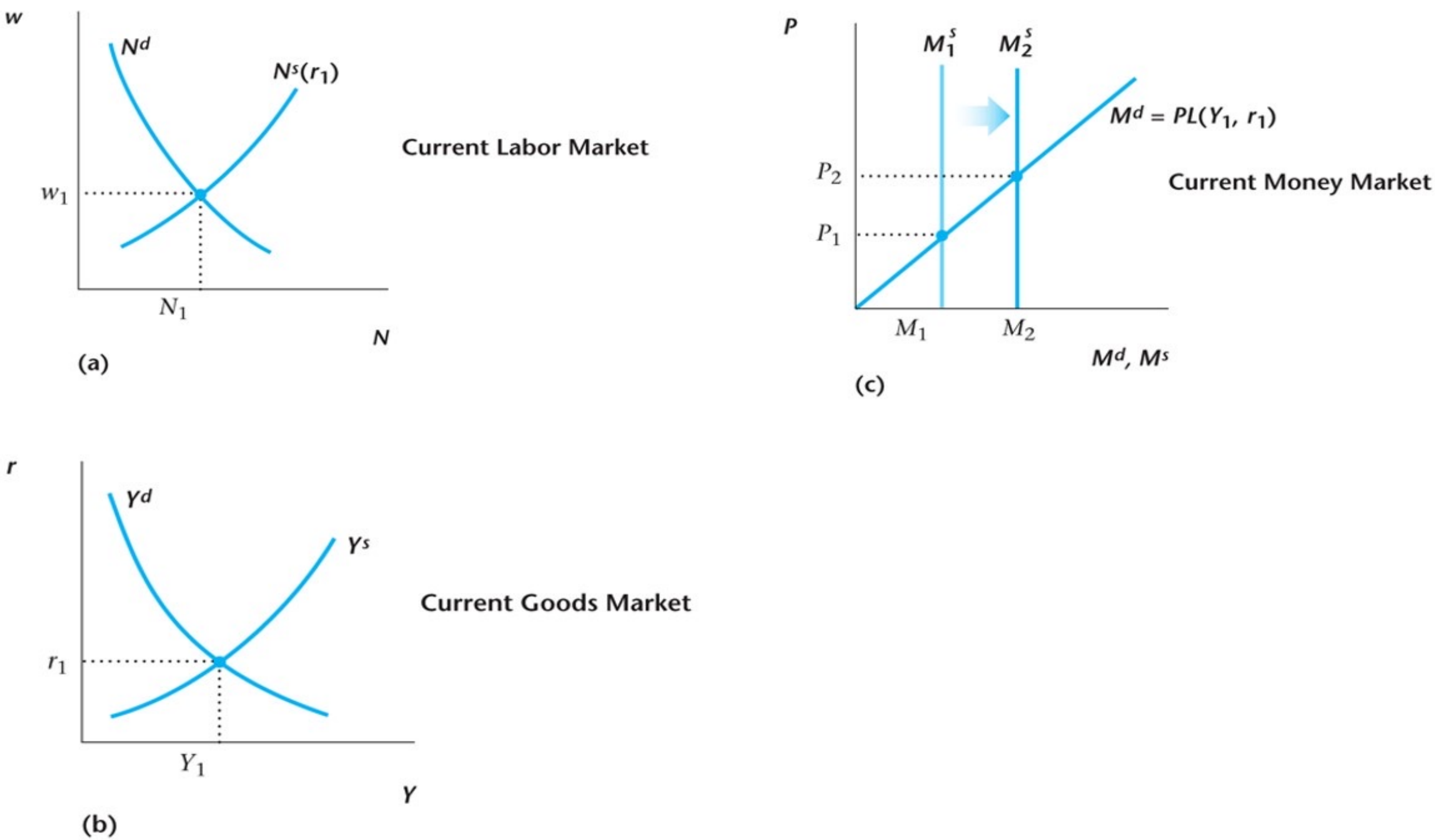
Figure 12.10 A Level Increase in the Money Supply in the Current Period



The Neutrality of Money

* In the monetary intertemporal model, a level increase in the money supply increases the price level and the nominal wage in proportion to the money supply increase, but has no effect on any real macroeconomic variable.

Figure 12.11 The Effects of a Level Increase in *M*—The Neutrality of Money



Shifts in Money Demand

- These shifts are important for how monetary policy should be conducted.
- Shifts in the demand for money that occur within a day, week or month (the very short run) are a critical for the central bank.

Figure 12.12 A Shift in the Supply of Credit Card Services

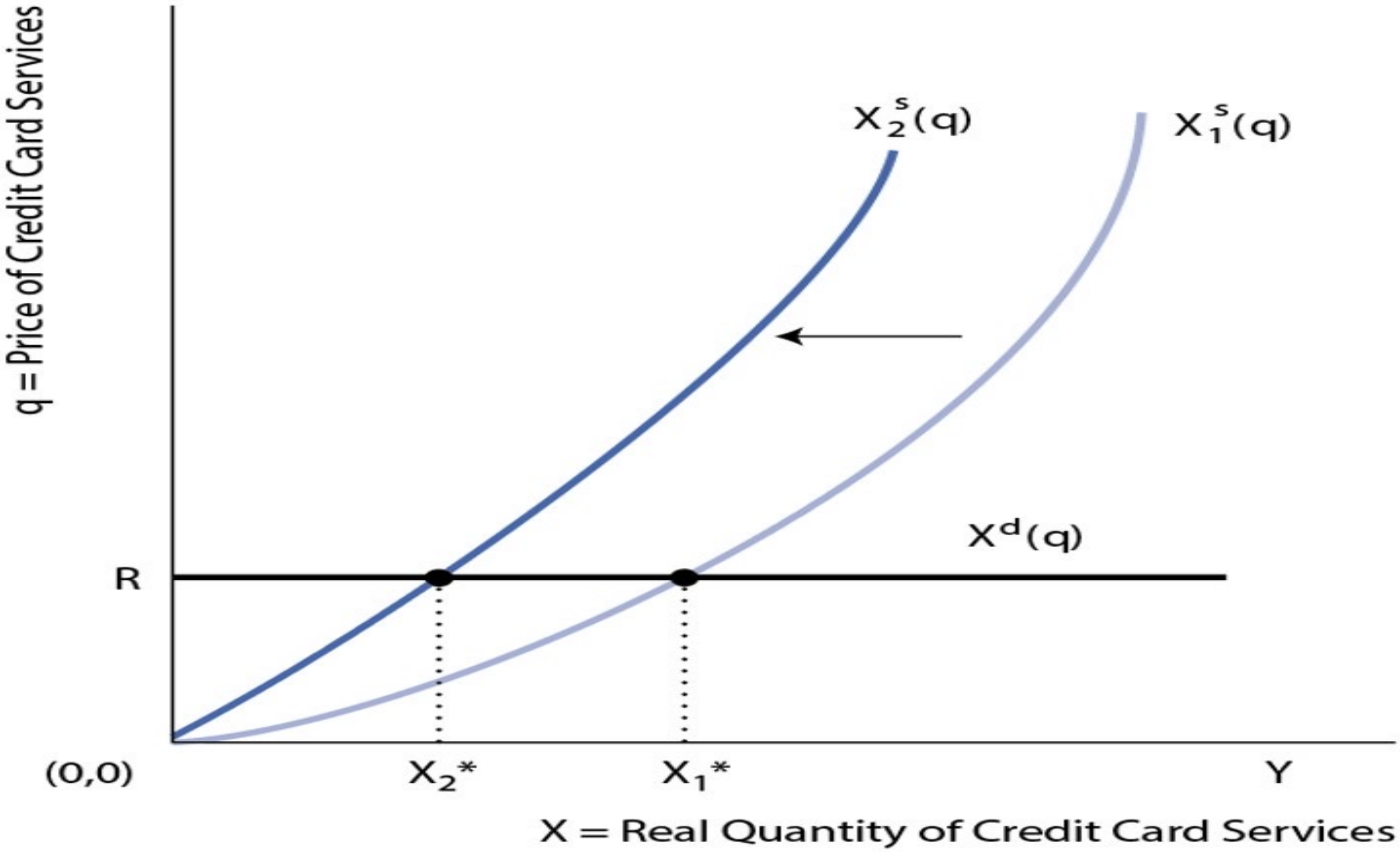


Figure 12.13 A Shift in the Demand for Money

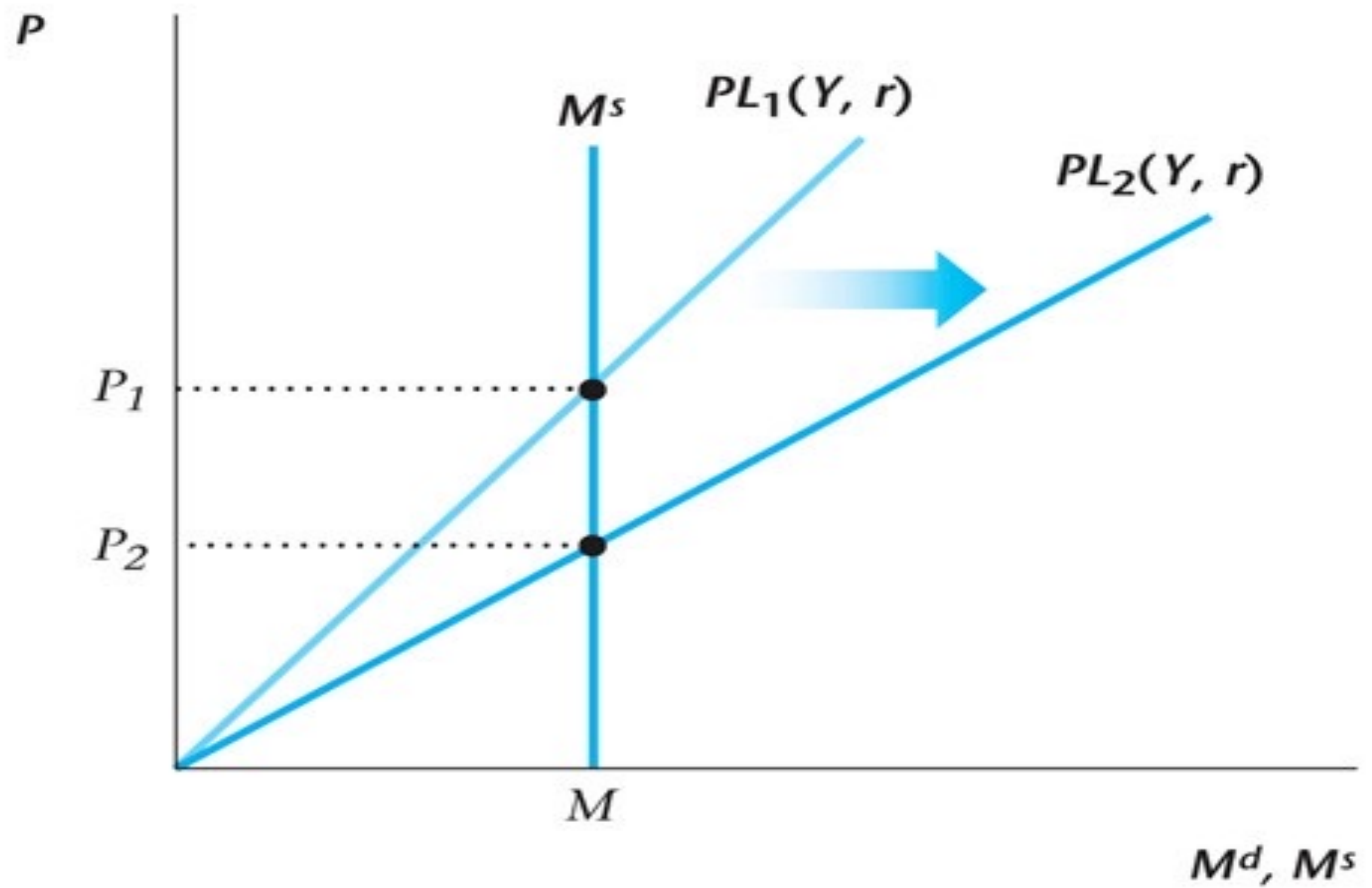
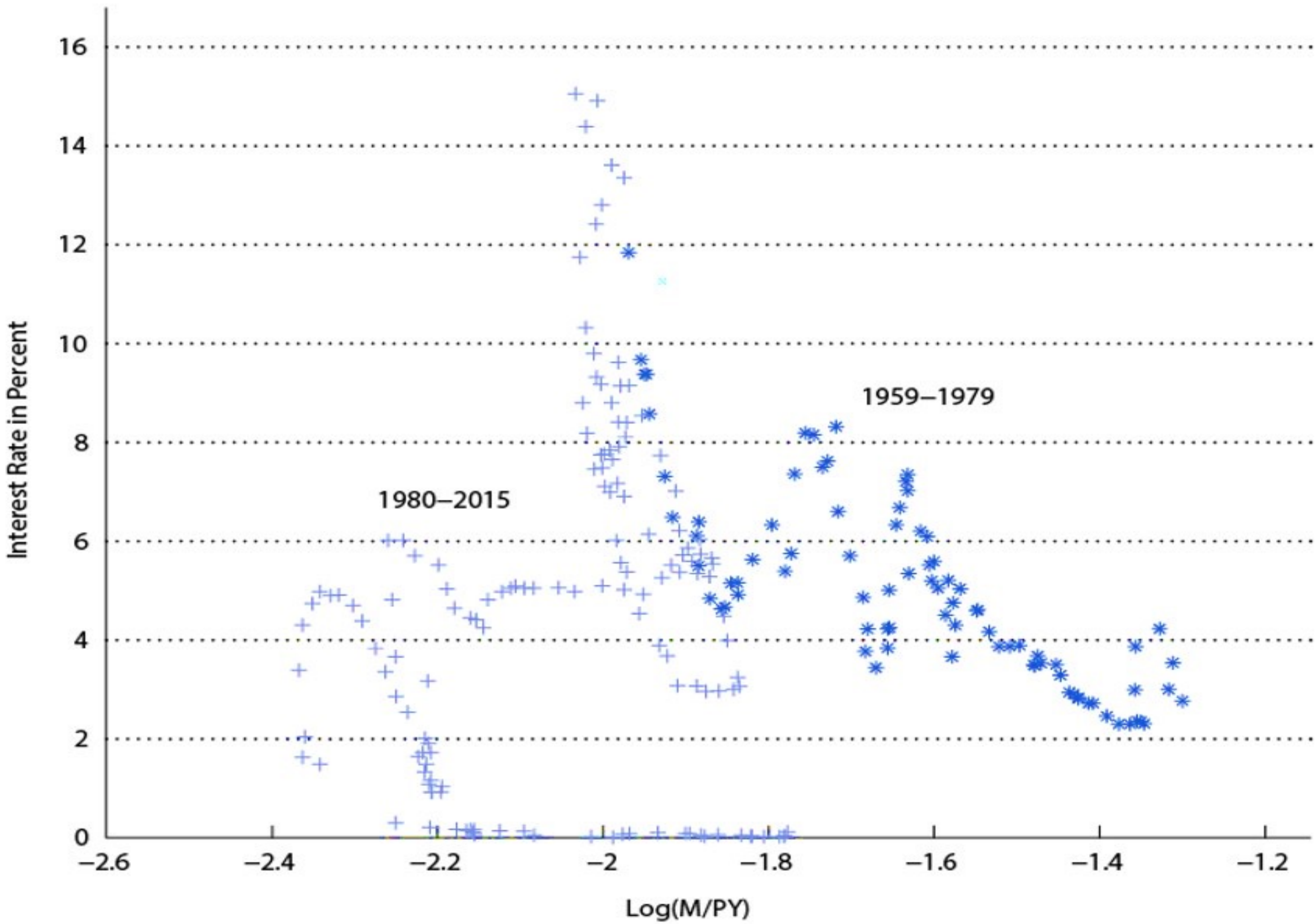


Figure 12.14 Instability in Money Demand



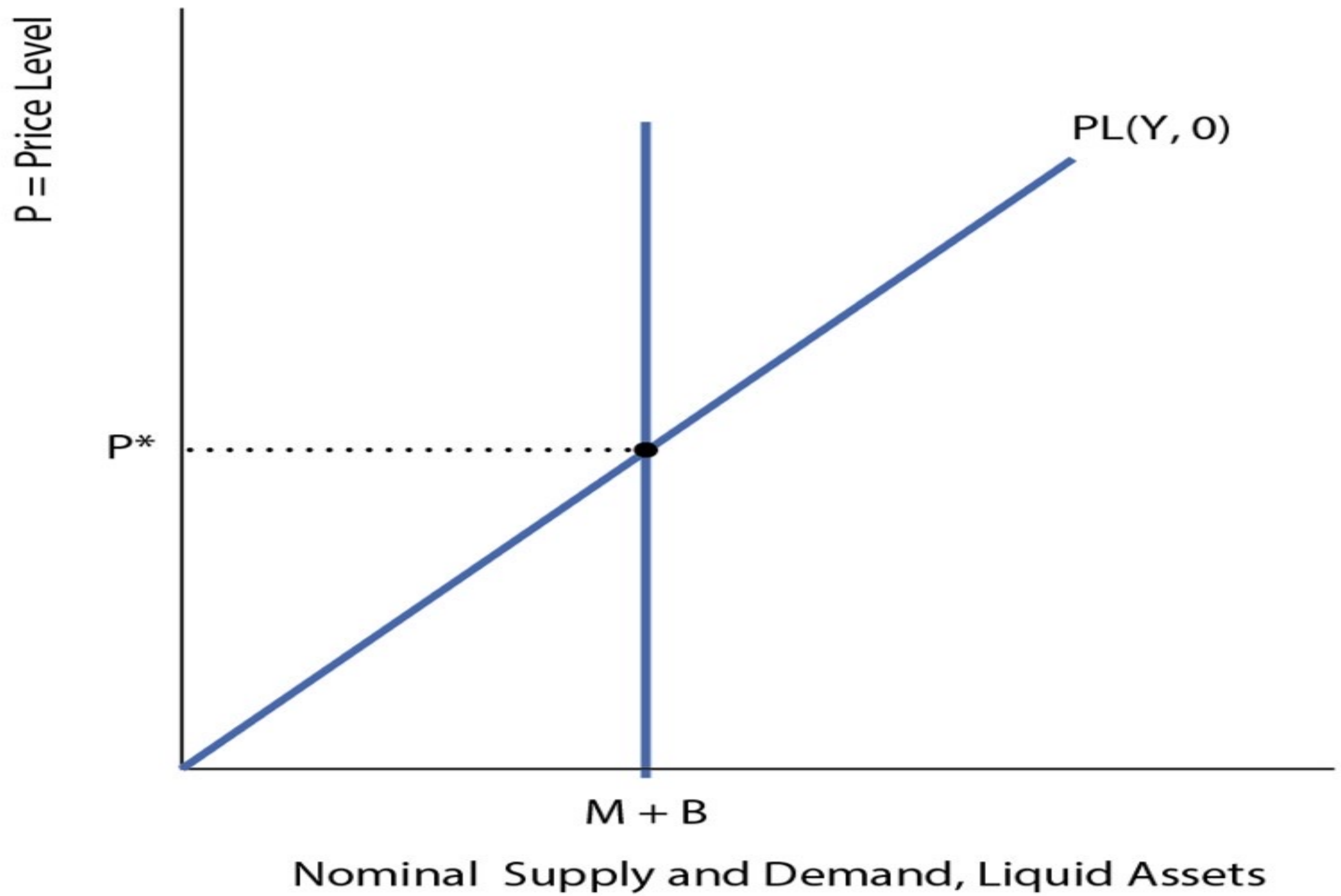
Sources of Shifts in the Demand for Money

- New information technologies.
- Changes in government regulations.
- Changes in the perceived riskiness of banks.
- Changes in hour-to-hour, day-to-day, week-to-week circumstances in the banking system.

The Liquidity Trap

- Typically thought that the nominal interest rate cannot go below zero (though in practice, not quite correct).
- There is a zero lower bound on the nominal interest rate.
- At the zero lower bound, money M and B become perfect substitutes.
- Liquidity trap: Open market purchases of B by the central bank will not matter.

Figure 12.15 A Liquidity Trap



Unconventional Monetary Policy

- Since conventional monetary policy – open market operations – does not work in a liquidity trap, central bank can resort to (among other policies):
  - Quantitative easing – purchases of long-maturity government debt and private assets.
  - Negative nominal interest rates – going below zero to the *effective lower bound*, by charging banks fees on reserve accounts with the central bank.